

3. The following table contains data about four of the planets that orbit the Sun.

	Distance from Sun (million km)	Surface temperature (°C)	Diameter (units)	Mass (units)	Gravitational field strength, <i>g</i> (N/kg)
Earth	150	15	10	60	10.0
Mars	230	−65	5	6	3.7
Saturn	1 435	−140	100	5 700	9.0
Venus	108	464	10	49	9.0

(adapted from planetary fact sheet NASA)

Use only information from the table to answer the following questions.

- (a)

Underline the correct word in each of the sentences below.

[3]

(i)

The planet nearest the Sun is (**Earth** / Mars / Saturn / Venus).

(ii)

The coldest planet is (**Earth** / Mars / Saturn / Venus).

(iii)

The biggest planet is (**Earth** / Mars / Saturn / Venus).
- (b)

(i)

State the gravitational field strength on Venus.

[1]

..... N/kg

(ii)

Use the equation:

weight = mass × gravitational field strength on Venus

to calculate the weight of an object of mass 80 kg on Venus.

[2]

weight = N

(iii)

Name a planet on which an object of mass 80 kg would have the same weight as on Venus.

[1]

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(c) One student, Ceri, says that planets with the same diameter have the same mass.
Explain whether you agree with this statement.

[2]

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